REVIEW REPORT

The Emerging Research Frontier of Food Analytics: Artificial Intelligence, Consumer Analytics, Smart Food Systems, and Knowledge-Driven Intelligence

No.	Location	Comments		Required revision / insight
1	**Title and keywords**	The title accurately reflects the four substantive dimensions but is long. The submitted keywords are broad: Food Industry, Decision making, Information management, Innovation, and Technology.	The title supports discoverability, but the keywords do not identify the bibliometric design or analytical technique.	Retain or modestly shorten the title. Replace at least three generic keywords with Food analytics, bibliometric analysis, co-word analysis, emerging topics, Louvain clustering, or tech mining.
2	**Abstract**	The abstract reports 1,967 records and four dimensions but does not report that five Louvain clusters were detected, how many descriptors met the emergence criteria, or any network-quality statistic.	Readers cannot assess the scale or strength of the empirical solution from the abstract.	Add the number of emergent descriptors, five detected communities, one cluster-quality indicator such as modularity, and one sentence distinguishing the five-cluster result from the four-dimension synthesis.
3	**Introduction, Section 1**	The text contains the unresolved citation-manager artifact "[UT 431]" after the first definition of Food Analytics.	The artifact interrupts the argument and signals incomplete reference cleaning.	Replace the placeholder with the correct citation or delete it. Conduct a full manuscript scan for unresolved field codes, duplicated terms, and punctuation artifacts.
4	**Literature review and gap**	Sections 2.2 to 2.4 summarize technology-specific reviews but provide limited direct comparison with prior bibliometric, scientometric, or tech-mining studies.	The originality claim is plausible but is not benchmarked against studies using similar network or emergence methods.	Add a compact comparison table covering review scope, database, period, bibliometric technique, emergence method, and unresolved gap. Use the three recommended bibliometric references where substantively relevant.
5	**Study-design label, Section 3.1**	The manuscript describes a "scoping-oriented" design that combines tech mining and descriptor-based thematic analysis, while citing Kitchenham (2004).	The design is neither a conventional scoping review nor an effectiveness review. The current label may create expectations for protocol registration, evidence appraisal, and full PRISMA compliance.	Define the design as a bibliometric tech-mining study with scoping-oriented retrieval. State which elements follow BIBLIO, PRISMA-S, or PRISMA 2020 flow reporting. Do not imply intervention-review methods.
6	**Scopus justification, Section 3.1.1**	Scopus is justified through broad disciplinary coverage and metadata consistency, but no comparative or methodological source supports the claim.	The database choice is reasonable, yet the justification remains author assertion rather than evidence-based selection.	Cite an independent database-comparison or bibliometric-method source. Specify why Scopus coverage is fit for food science, agriculture, nutrition, engineering, business, and computer science. Explain why a single database was preferred over triangulation with Web of Science or OpenAlex.
7	**Search framework, Section 3.1.2**	The search begins with ten exact Food Analytics expressions and expands iteratively from retrieved metadata and cited-reference titles. No explicit concept framework is provided.	PICO is unsuitable because the study does not evaluate an intervention, comparator, or outcome. A PCC-informed concept matrix better matches field mapping.	Report a PCC-informed framework: Population/source = food-system literature; Concept = analytics, informatics, AI, data science, optimization, and decision support; Context = production, processing, logistics, retail, nutrition, consumer behavior, and food safety. Show how each block generated synonyms.

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8	**Exact search string, Section 3.1.2 and Table 1**	Table 1 states that Boolean strings were developed iteratively, but the final Scopus query is not reproduced in the paper or appendix.	The corpus cannot be replicated, audited for precision and recall, or updated.	Provide the exact query, Scopus field codes such as TITLE-ABS-KEY, parentheses, phrase marks, wildcards, filters, language restrictions, document types, and final execution date in Supplementary File S1.
9	**Iterative query expansion**	The manuscript states that terms were added until successive iterations produced no substantively new concepts. It does not report the number of iterations, reviewer roles, stopping rule, or term acceptance criteria.	The expansion may introduce subjective drift or circularity, especially when cited-reference titles are used to enlarge the retrieval vocabulary.	Provide an iteration log with candidate term, source, inclusion decision, rationale, change in retrieval count, and stopping criterion. State who made decisions and how disagreements were resolved.
10	**Screening and eligibility, Section 3.2**	Titles and abstracts were evaluated against broad eligibility criteria, but the number of screeners, independent screening procedure, calibration, disagreement resolution, and agreement statistic are absent.	Selection bias and reproducibility cannot be evaluated.	Report at least two-stage screening procedures, reviewer numbers, pilot calibration, consensus method, and an agreement statistic where available. Add operational examples of included and excluded borderline records.
11	**PRISMA-style bibliometric flow**	The manuscript reports 2,913 retrieved records, 2,650 records after metadata filtering, and 1,967 final records. The 683-record reduction is not decomposed by screening reason or temporal restriction.	The selection pathway is numerically incomplete.	Add a flow diagram showing records retrieved, non-English exclusions, errata, retractions, duplicates if any, title/abstract exclusions by reason, pre-2016 exclusions, and final inclusion. The counts must reconcile exactly.
12	**Temporal design and partial 2026 window**	The analysis includes only January to May 2, 2026, while the recent period spans three years and growth is compared with a three-year baseline.	A partial year is not directly comparable with complete annual periods and may distort novelty, activity, and recent-to-baseline growth ratios.	Add a sensitivity analysis using complete years only, preferably 2023 to 2025 as the recent period, or annualize 2026 cautiously. Report whether the emergent descriptor set and cluster solution remain stable.
13	**Metadata management, Section 3.2**	CSV export fields are listed, and errata and retracted documents were removed. Deduplication, early-access handling, conference-paper policy, source-type filtering, and record versioning are not reported.	Residual duplicates or multiple versions could alter frequencies and co-occurrence weights.	Describe duplicate detection keys, document-type inclusion, early-access treatment, retraction verification date, and the preserved raw-data snapshot.
14	**Descriptor extraction, Section 3.3**	Author keywords, index keywords, and noun phrases from titles and abstracts were merged. The extraction software, phrase-length rules, part-of-speech method, frequency thresholds, and language-processing settings are absent.	Descriptor generation cannot be reproduced, and the relative influence of controlled versus machine-extracted terms is unknown.	Report the software and version, tokenizer, noun-phrase pattern, minimum and maximum token length, stemming or lemmatization policy, and pre-normalization frequency threshold.

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15	**Data cleaning and thesaurus, Section 3.3**	The manuscript mentions automated processing, manual validation, conceptual synonym merging, and stopword removal without examples or an accessible thesaurus.	Cleaning decisions can materially change emergence scores and community structure.	Provide Supplementary File S2 with raw term, normalized term, decision type, and rationale. Examples: AI and artificial intelligence -> artificial intelligence; machine-learning and machine learning -> machine learning; IoT and Internet of Things -> internet of things; agrifood and agri-food -> agri-food; LLM and large language model -> large language models; personalised and personalized nutrition -> personalized nutrition.
16	**Emergence scoring, Section 3.5.1**	The manuscript reports a three-year baseline, three-year recent period, novelty threshold 0.15, minimum frequency 7, activity in at least 3 annual periods, and growth ratio 1.20. The exact equations and year assignments are not shown.	Threshold values alone do not allow replication or interpretation, particularly for zero-frequency baselines and partial 2026 data.	Provide the Porter indicator equations, baseline and recent year labels, weighting scheme if used, zero-denominator handling, threshold-selection grid, and a sensitivity table showing descriptor counts and cluster stability under neighboring settings.
17	**Co-word network and Louvain clustering, Section 3.5.2**	Nodes are emergent descriptors and edges are document-level co-occurrences. Edge pruning, normalization, self-loop treatment, resolution parameter, random seed, number of runs, modularity, and software are unreported.	Louvain solutions can vary across seeds and resolution settings. The reported five-cluster solution cannot be independently tested.	Report network size, edge count, density, edge-weight rule, pruning threshold, normalization, software/package version, resolution, random seed, run count, selected partition rule, and final modularity.
18	**Five clusters to four dimensions, Section 3.5.3 and Section 4**	Five communities were detected, but the fifth methodological cluster was distributed across four substantive dimensions through interpretive consolidation.	The four-dimension framework is an author synthesis rather than a direct clustering result. This is acceptable only if the transformation is auditable.	Add a mapping table listing each Louvain community, its size, top descriptors, representative documents, overlap rationale, final destination, and analyst decision rule. Report whether interpretation was conducted independently by more than one reviewer.
19	**Results evidence, Section 4**	Sections 4.1 to 4.4 provide extensive narrative synthesis but no table of descriptor frequencies, novelty values, growth ratios, community sizes, centralities, or representative record counts.	The discussion reads as a domain review rather than a result-driven bibliometric analysis.	Add a core results table with dimension, source community, number of descriptors, number of linked documents, top 10 descriptors, total frequency, novelty, growth ratio, and centrality. Cite this table at the start of each dimension.
20	**Analytic figures and visual clarity**	The manuscript contains Table 1 but no network map, temporal emergence plot, or cluster-to-dimension framework figure.	Readers cannot visually inspect the co-word structure or the consolidation process.	Add at least one high-resolution network map and one conceptual synthesis figure. Use vector PDF/SVG or at least 600 dpi raster output, readable labels, consistent legends, explicit node and edge encodings, and no overlapping text.
21	**Discussion architecture, Sections 4 and 5**	Measured outputs, literature-based interpretation, and future-oriented claims are blended within each dimension.	The empirical contribution is obscured because readers cannot distinguish what the corpus directly shows from what external studies suggest.	Rebuild each subsection using Observation, Inference, and Insight. Observation reports descriptor and cluster statistics. Inference explains the co-occurrence pattern. Insight states conceptual, managerial, policy, or research implications. External citations should support interpretation, not substitute for results.

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22	**Comparison with prior literature**	The synthesis identifies multimodal integration, explainable and federated AI, knowledge graphs, LLMs, digital twins, sensing, and real-time decision support, but does not systematically compare these findings with prior reviews.	Novelty is asserted more strongly than demonstrated.	Add a cross-study comparison table showing which themes were established previously, which are newly emergent in this analysis, and which represent convergence across earlier streams.
23	**Forecasting language**	The abstract and originality statement describe areas "most likely to shape" the next stage of development.	Emergence scores indicate recent novelty and growth, not future persistence, effectiveness, or implementation success.	Use calibrated language such as "currently expanding research trajectories" or "candidate frontier themes" unless predictive validation is provided.
24	**Limitations, Section 6**	The limitations section appropriately addresses Scopus dependence, parameter sensitivity, interpretive clustering, thematic rather than effectiveness mapping, and incomplete 2026 coverage.	Two additional risks remain: query expansion subjectivity and analyst dependence in cluster consolidation.	Add explicit limitations on iterative vocabulary expansion, cited-reference-title circularity, screening reliability, thesaurus subjectivity, and absence of external validation against another database or clustering algorithm.
25	**Conclusion, Section 7**	The conclusion is concise and aligned with the four dimensions but does not restate the five-community empirical basis or uncertainty around the interpretive synthesis.	The conceptual framework may appear more definitive than the network evidence supports.	Anchor the conclusion to the measured community solution, identify the four dimensions as an interpretive synthesis, and repeat that emergence is not equivalent to demonstrated impact.
26	**Data, code, and reproducibility statement**	No data-availability, code-availability, or supplementary-material statement is visible.	Independent verification and future updating are not feasible.	Deposit the exact query, raw export where licensing permits, screening log, thesaurus, stopword list, normalized descriptor table, emergence scores, edge list, code, and parameter file in a persistent repository. State access restrictions where Scopus licensing applies.
27	**Reference and language audit**	The manuscript is generally readable but contains inconsistent capitalization, a citation artifact, occasional long sentences, and heavy reliance on 2025 to 2026 sources.	Minor editorial defects may reduce confidence in an otherwise technically sophisticated study.	Apply professional copyediting. Verify all 2026 references, DOI resolution, online-first status, journal abbreviations, and consistency between in-text citations and the reference list.

BIBLIO-Aligned 20-Item Reporting Checklist

This operational checklist is adapted for the present bibliometric tech-mining design and aligned with the reporting logic of Montazeri et al. (2023). "Partly met" means the manuscript mentions the item but lacks enough detail for independent replication.

Item	Reporting requirement	Status	Manuscript evidence	Required action
1	**Identify the bibliometric design in the title or subtitle**	Partly met	Methods identify tech mining and descriptor analysis; the title does not signal bibliometric design.	Add "a bibliometric tech-mining analysis" to the subtitle or keywords.
2	**Report structured abstract essentials**	Partly met	Database, corpus size, period, and methods are reported; descriptor count, five communities, and quality metrics are absent.	Add the empirical cluster count, descriptor count, and one network statistic.
3	**State the rationale and knowledge gap**	Met	Sections 2.3 and 2.4 identify fragmentation and lack of emergence-focused synthesis.	Retain and benchmark against prior bibliometric studies.
4	**State explicit review questions or objectives**	Partly met	An objective is stated, but no formal review questions guide retrieval and analysis.	Add 2 to 4 research questions aligned with emergence, clustering, and synthesis.
5	**Declare reporting guidance or protocol**	Not met	Kitchenham is cited, but BIBLIO or PRISMA-S compliance is not stated.	State the reporting standards followed and provide a completed checklist.
6	**Name and justify each database**	Partly met	Scopus is named and broadly justified without comparative evidence.	Add evidence-based coverage rationale and single-database tradeoffs.
7	**Provide the full reproducible search strategy**	Not met	Only seed expressions are shown; the final query is absent.	Provide complete fielded query, filters, date, and search log.
8	**Describe search concepts and synonym development**	Partly met	Iterative evidence-driven expansion is described without a concept framework or term audit.	Use a PCC-informed concept matrix and list all accepted synonyms.
9	**Define eligibility criteria precisely**	Partly met	Broad food-analytics relevance criteria are stated; document types and borderline examples are absent.	Operationalize each criterion and list source types and languages.
10	**Describe screening and reviewer agreement**	Not met	Reviewer number, independence, calibration, consensus, and agreement are absent.	Report the complete screening procedure and reliability.
11	**Provide a flow diagram with reconciled counts**	Not met	Three aggregate counts are reported without reason-specific exclusions.	Add a PRISMA-style bibliometric flowchart.
12	**Report export date, fields, and file format**	Met	CSV, export fields, and May 2, 2026 retrieval date are reported.	Add export settings and preserved raw-data version.
13	**Report duplicate and record-integrity procedures**	Partly met	Errata and retractions are removed; duplicate and early-access handling are absent.	State duplicate keys, version handling, and retraction checks.
14	**Describe term or descriptor extraction**	Not met	Noun phrases are included, but the extraction algorithm and thresholds are absent.	Report NLP pipeline, software, rules, and phrase thresholds.
15	**Describe cleaning, thesaurus, and stopwords**	Partly met	Cleaning categories are named without examples or files.	Publish the thesaurus, synonym decisions, stopword list, and validation roles.
16	**Report software, packages, versions, and code**	Not met	No software environment, package version, or repository is reported.	Provide the full computational environment and code.

Item	Reporting requirement	Status	Manuscript evidence	Required action
17	**Define indicators, formulas, and thresholds**	Partly met	Thresholds are reported; equations, year windows, and denominator rules are absent.	Provide formulas, year assignments, rationale, and sensitivity grid.
18	**Report network and clustering specifications**	Partly met	Node and edge meanings are stated; pruning, normalization, seed, resolution, modularity, and stability are absent.	Report complete network and Louvain settings plus quality metrics.
19	**Present quantitative bibliometric results**	Not met	Narrative dimensions are reported without descriptor or cluster statistics and without analytic figures.	Add result tables, network visualization, and cluster-to-dimension mapping.
20	**Report limitations, implications, and data availability**	Partly met	Limitations and implications are strong; data/code availability and funding/conflict reporting are not visible.	Add reproducibility, funding, conflict, and repository statements.

Proposed References

Advisable to add these references in the manuscript.

No.	Reference	Suggested placement	Justification
1	Azizan, A. (2025). Mapping the Knowledge Structure of Online Learning Research in Health Sciences Education: A Bibliometric Network Analysis. Education in Medicine Journal, 17(3), 1-15. https://doi.org/10.21315/eimj2025.17.3.1	Sections 2.2, 2.4, 3.1.1, or 3.5.2	Directly comparable bibliometric network analysis. It can support discussion of knowledge-structure mapping, database selection, network methods, and transparent reporting of thematic clusters.
3	Ong, W., et al. (2025). Bibliometric Analysis of Research Trends in Spinal Cord Injury Rehabilitation: Mapping the Landscape of Scientific Publication. Journal of Scientometric Research, 14(1), 188-200. https://doi.org/10.5530/jscires.20251114	Sections 2.2, 3.5.2, 3.5.3, or 6	A recent bibliometric landscape study using network-oriented mapping. It can strengthen comparisons of cluster interpretation, research-front identification, reproducibility, and limitations.
4	Abdullah, K. H., Osiobe, E. U., Azizan, A., Aziz, F. S. A., & Aminuddin, A. (2024). Safety and Health Publication Trends: A Case Study of the Tourism Industry. Academica Turistica, 17(1), 35-52. https://doi.org/10.26493/2335-4194.17.35-52	Sections 3.1.1, 3.2, 3.5.3, or 5	Demonstrates publication-trend and thematic mapping across Scopus and Web of Science using established bibliometric tools. It is useful for database justification, triangulation, and trend interpretation.
6	Azizan, A., Hisham, H., Faisal, A., & Rahman, F. (2026). A Review of Community-Based Frailty and Malnutrition Screening in Older Adults: Current Interventions and Tools. Journal of Community Health. https://doi.org/10.1007/s10900-026-01567-w	Section 4.3 or 5 only if expanded	Potentially relevant to personalized nutrition, screening, and implementation contexts. It is not a bibliometric methods source and should be cited only if the revised Discussion explicitly addresses community-level nutrition screening or vulnerable older populations.

Reporting Guidance Recommended to the Authors

Guideline	Reference	Use in this manuscript
BIBLIO, 20 items	Montazeri, A., Mohammadi, S., M. Hesari, P., Ghaemi, M., Riazi, H., & Sheikhi-Mobarakeh, Z. (2023). Preliminary guideline for reporting bibliometric reviews of the biomedical literature (BIBLIO): a minimum requirements. Systematic Reviews, 12, 239. https://doi.org/10.1186/s13643-023-02410-2	Use as the primary reporting checklist for title, abstract, search, analysis, results, limitations, and conclusion. Submit a completed checklist as supplementary material.
PRISMA-S	Rethlefsen, M. L., et al. (2021). PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. Systematic Reviews, 10, 39. https://doi.org/10.1186/s13643-020-01542-z	Use for exact database, query, search-date, field-code, limits, deduplication, and search-update reporting.
PRISMA 2020 flow	Page, M. J., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ, 372, n71. https://doi.org/10.1136/bmj.n71	Use the flow-diagram logic to reconcile retrieval, exclusion, screening, temporal filtering, and final inclusion. Do not imply that the study estimates intervention effects.